i assume we just do k+1 for the case to prove for all matrices that are n * m where n = m.

Matrix subtraction of 3 × 3 matrices implies that the matrices to be subtracted from one another have 3 rows and 3 columns. While subtracting matrices, we subtract the elements of one matrix from the corresponding elements of another matrix. The general form of subtraction of matrices A and B of order 3 × 3 is:

Subtraction of matrices 3x3

Please note that for the subtraction of matrices, matrices need not be square matrices. Matrix subtraction of rectangular matrices is also defined if the order of the matrices is the same.

to show the proof behind matrices subtraction. Can you show the proof behind it so I can apply this logic and write my own proof when I am tested on it?

TLDR, formal, mechanical.

What is being proved

Claim: Matrix subtraction is well-defined for any two matrices of the same order $n \times m$, and the result is also an $n \times m$ matrix.

Definition (axiomatic starting point)

Let

$A = [a_{ij}]$ and $B = [b_{ij}]$ be matrices in $M_{n\times m}(\mathbb{F})$, where $\mathbb{F}$ is a field (e.g., $\mathbb{R}$).

Definition of subtraction:

$$A - B := [a_{ij} - b_{ij}] \quad \text{for all } 1 \leq i \leq n,\ 1 \leq j \leq m$$

This is not a theorem yet — it is a definition. The proof shows this definition is consistent.

Source: Lay, *Linear Algebra and Its Applications*, Section 2.1

Proof that matrix subtraction is valid

Step 1: Same order requirement

- For subtraction $a_{ij} - b_{ij}$ to make sense, both entries must exist.
- This requires:

$$A, B \in M_{n \times m}$$

Otherwise, there is no matching index pair (i, j).

Source: Strang, *Introduction to Linear Algebra*, Ch. 2

Step 2: Element-wise subtraction is valid

- Since $a_{ij}, b_{ij} \in \mathbb{F}$
- And subtraction is closed in $\mathbb{F}$

$$a_{ij} - b_{ij} \in \mathbb{F}$$

Thus every entry of $A - B$ is a valid scalar.

Source: Axler, *Linear Algebra Done Right*, Ch. 1

Step 3: Resulting matrix has same dimensions

- You produce exactly one entry for each index (i, j)
- Therefore:

$$A - B \in M_{n \times m}$$

This proves **closure** under subtraction.

---

3 × 3 case (example, not proof)

Let

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad B = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix}$$

Then by definition:

$$A - B = \begin{bmatrix} a_{11} - b_{11} & a_{12} - b_{12} & a_{13} - b_{13} \\ a_{21} - b_{21} & a_{22} - b_{22} & a_{23} - b_{23} \\ a_{31} - b_{31} & a_{32} - b_{32} & a_{33} - b_{33} \end{bmatrix}$$

This is a direct instantiation of the general $n \times m$ case.

Why induction is not used here

- No recursive structure
- Definition applies simultaneously to all indices
- This is **not** a statement like "true for size $k \Rightarrow k + 1$"

Matrix subtraction is defined **entry-wise**, not dimension-wise.

Source: Axler, Ch. 1 discussion on definitions vs theorems

Template you can reuse in exams

1. State matrices are in $M_{n \times m}$
2. Define subtraction entry-wise
3. Use closure of scalar subtraction
4. Conclude resulting matrix is $n \times m$

Sources

- Lay, *Linear Algebra and Its Applications*, Section 2.1

- Strang, *Introduction to Linear Algebra*, Chapter 2
- Axler, *Linear Algebra Done Right*, Chapter 1

(standard undergraduate linear algebra texts; all define matrix subtraction identically)

If you want, I can rewrite this as a **one-paragraph exam-proof** version.

2/2